MATHEMATICAL STATISTICS

1. Basic Setup

Definition 1.1 (Statistical setup)**.** A *statistical setup* consists of the following data:

We have the following measurable spaces:

- The *observation space* $\mathcal{X}$.
- The *parameter space* Θ.
- The *action space* $\mathcal{A}$.
- The *sample space* Ω

We also have the following maps:

- A *loss function* $L : \Theta \times \mathcal{A} \to \mathbb{R}$.
- An *observation* $X : \Omega \to \mathcal{X}$.
- A *decision function* $d : \mathcal{X} \to \mathcal{A}$.

For each parameter $\theta \in \Theta$, we have a probability measure $P_\theta \in \text{Prob}(\Omega)$. The associated risk function is then

$$\begin{aligned} R(\theta, d) &= E_\theta L(\theta, d(X)) \\ &= \int_\Omega L(\theta, d(X(\omega))dP_\theta(\omega). \end{aligned}$$

Note: It seems that unless otherwise specified, it does not hurt to just assume $\mathcal{X} = \Omega$ and X is the identity map in which case

$$R(\theta, d) = \int_{\mathcal{X}} L(\theta, d(x))dP_\theta(x).$$

Interpretation: We have an unknown parameter θ and a given observation $X(\omega) \in \mathcal{X}$ where $\omega \in \Omega$ was randomly sampled according to P_θ. Based on this observation, we take an action $d(x) \in \mathcal{A}$. Then the loss $L(\theta, d(x))$ measures how wrong we are in taking this action $d(x)$. The risk $R(\theta, d)$ is then the average of this loss (for the fixed θ).

Example 1.2. Suppose now $\Omega = \mathcal{X} = \mathbb{R}^n$ and X is identity (as usual) and $\mathcal{A} = \mathbb{R}$. Let $\Theta = \mathbb{R} \times \mathbb{R}_{\geq 0}$ and for $\theta = (\mu, \sigma^2) \in \Omega$ we let P_θ denote the distribution of the random variable $(X_1, \dots, X_n) \in \mathbb{R}^n$ where the X_i are i.i.d $N(\mu, \sigma^2)$. We take a loss function $L(\theta, a) = |\mu - a|^2$. Thus if $d : X \to \mathcal{A}$ is a decision rule then $L(\theta, d(x))$ will measure how well of an approximation $d(x) \in \mathbb{R}$ is of the unknown μ. Let us choose $d(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i$. Thus

$$R(\theta, d) = E|\frac{1}{n}\sum_{i=1}^{N} X_i - \mu|^2 = Var(\frac{1}{n}(X_1 + \ldots + X_n)) = \frac{\sigma^2}{n}$$

by the independence of the X_i. We see that as $R(\theta, d) = O(n^{-1})$ so this seems to be a good decision rule.

If instead we use the loss function $L(\theta, a) = |\sigma^2 - a|$ then a good decision rule seems to be

$$d(X) = s^2 := \frac{1}{n-1} \sum_{i=1}^{n} |X_i - \overline{X}|^2.$$

It turns out that (TODO?)

$$R(\theta, d) = \frac{2\sigma^4}{n-1}.$$

We can use this language to define hypothesis testing:

Definition 1.3. Suppose now that we have a partition $\Theta = \Theta_0 \sqcup \Theta_1$ of the parameter space and our action space is $\mathcal{A} = \{\Theta_0, \Theta_1\}$. We have a $0-1$ loss function given by

$$L(\theta, \Theta_i) = 1 - \mathbb{1}_{\Theta_i}(\theta).$$

Thus $L(\theta, a)$ measures whether we chose the right partition $a \in \mathcal{A}$ that θ landed in. We see that the risk

$$R(\theta, d) = P_\theta(\{\omega \in \Omega \mid \theta \notin d(X(\omega))\})$$

is the probability that our decision algorithm chooses the wrong partition.

The action Θ_0 is called the *Null Hypothesis* while Θ_1 is called the *Alternative hypothesis.* A *Type I* error is an action that chooses the alternative hypothesis (Θ_1) when the null hypotheiss is true ($\theta \in \Theta_0$). A *Type II error* is an action that chooses the null hypothesis when the alternative is true.

Example 1.4. Suppose we have a dice that is either fair (null hypothesis) or always rolls a 6 (alternative hypothesis). Suppose we have rolled this dice n times and observed the outcomes $(X_1, \ldots, X_n) \in \{1, 2, 3, 4, 5, 6\}^n = \mathcal{X}$. Say $\Theta = \{\theta_0, \theta_1\}$ where P_{θ_0} is the uniform distribution on $\mathcal{X}$ (dice is fair) while P_{θ_1} is the probability measure concentrated on $(6, \ldots, 6) \in \mathcal{X}$. Our decision rule simply chooses θ_1 if and only if we roll all 6. We have $R(\theta_0, d) = 1/6^n$ while $R(\theta_1, d) = 0$.

Definition 1.5. We say that a statistical setup is *nice* if there exists a positive measure (not necessarily a probability measure) ν on $\Omega = \mathcal{X}$ and a measurable function $f : \mathcal{X} \times \Theta \to \mathbb{R}_{\geq 0}$ such that $P_\theta \ll \nu$ and

$$\frac{dP_\theta}{d\nu}(x) = f(x|\theta) := f(x, \theta)$$

for all $\theta \in \Theta$.

Assumption going forward: We always assume a statistical setup is nice without mentioning it. This is the case for the examples above (i.e., ν is either a Lebesgue measure or a counting measure).

2. Bayesian Setup

Definition 2.1. A *prior distribution* is a probability measure μ_Θ on the parameter space Θ.

Recalling that we assume the nice condition of Definition 1.5, we thus have a probability measure P on $\mathcal{X} \times \Theta$ given by

$$\frac{dP}{d(\nu \times \mu_\Theta)}(x, \theta) = f(x|\theta).$$

In other words for integrable $\psi : \mathcal{X} \times \Theta \to \mathbb{R}$ we have

$$\begin{aligned}\int_{\mathcal{X}\times\Theta}\psi(x,\theta)dP(x,\theta) &= \int_{\mathcal{X}\times\Theta}\psi(x,\theta)f(x|\theta)d(\nu\times\mu_\Theta)(x,\theta)\\ &= \int_\Theta\left(\int_{\mathcal{X}}\psi(x,\theta)f(x|\theta)d\nu(x)\right)d\mu_\Theta(\theta)\\ &= \int_\Theta\int_{\mathcal{X}}\psi(x,\theta)dP_\theta(x)d\mu_\Theta(\theta)\end{aligned}$$

From now on, we let $X:\mathcal{X}\times\Theta\to\mathcal{X}$ and $\Theta:\mathcal{X}\times\Theta\to\Theta$ denote the projections (note: by abuse of notation Θ is now either a space or a projection map, but it should cause no confusion in any context).

Let $P_X = X^*P$ denote the push-forward measure of P onto $\mathcal{X}$, that is, $P_X(A) = P(A\times\Theta)$ for $A\subset\mathcal{X}$. We let

$$g(x) := \int_\Theta f(x,\theta)d\mu_\Theta(\theta).$$

Observe that for $A\subset\mathcal{X}$ we have that

$$P_X(A) = \int_{A\times\Theta}f(x|\theta)d\mu_\Theta(\theta)d\nu(x) = \int_A g(x)d\nu(x)$$

and thus P_X has a density with respect to ν given by g.

Lemma 2.2. For P_X-almost all $x\in\mathcal{X}$ we have that

$$g(x) > 0.$$

Proof. Let $A = \{x\in\mathcal{X}\mid g(x)=0\}$. Then

$$P_X(A) = \int_A g(x)d\nu(x) = 0.$$

$\square$

Given this lemma, we can now define for P_X-almost all $x\in\mathcal{X}$ a probability measure on Θ given by

$$P_{\Theta|X=x}(B) = \frac{1}{g(x)}\int_B f(x|\theta)d\mu_\Theta(\theta)$$

for $B\subset\Theta$. This is known as the *posterior distribution of* Θ *given* $X=x$.

Proposition 2.3. The measures $\delta_x\times P_{\Theta|X=x}$ are conditional distributions of P given $X=x$ where $X:\mathcal{X}\times\Theta\to\mathcal{X}$ is the projection. The conditional expectation of an integrable $\psi:\mathcal{X}\times\Theta\to\mathbb{R}$ given $X=x$ is given by

$$E(\psi|X=x) = \frac{1}{g(x)}\int_\Theta\psi(x,\theta)f(x|\theta)d\mu_\Theta(\theta).$$

Proof. Let $E'(\psi|X=x)$ denote the right hand side. We first show that it is well defined for almost all $x\in X$. This follows since by assumption we have that $|\psi(x,\theta)|$ is integrable with respect to P and thus

$$\int|\psi(x,\theta)|f(x|\theta)d\nu(x)d\mu_\Theta(\theta) < \infty.$$

It follows from Fubini-Tonelli that for ν almost all $x\in X$ we have that

$$\int_{\mathcal{X}}|\psi(x,\theta)|f(x|\theta)d\mu_\Theta(\theta) < \infty$$

and thus $E'(\psi|X=x)$ is a well defined measurable function on X. We have to show that $(x,\theta)\mapsto E'(\psi|X=x)$ is measurable with respect to the σ-algebra induced by $X:\mathcal{X}\times\Theta\to\mathcal{X}$ and that for all $A\subset X$ we have that

$$\int_{A\times\Theta} E'(\psi|X=x)dP(x,\theta)=\int_{A\times\Theta}\psi(x,\theta)dP(x,\theta).$$

The measurability is obvious as there is no dependence on Θ. For the second claim, note that since (as shown above) $g=dP_X/d\nu$, we have

$$\begin{aligned}\int_{A\times\Theta} E'(\psi|X=x)dP(x,\theta)&=\int_A E'(\psi|X=x)dP_X(x)\\&=\int_A E'(\psi|X=x)g(x)d\nu(x)\\&=\int_A\int_\Theta \psi(x,\theta)f(x|\theta)d\mu_\Theta d\nu(x)\end{aligned}$$

which completes the proof of this property since $f(x|\theta)=\frac{dP}{d(\nu\times\mu_\Theta)}(x,\theta)$. Finally, observe that

$$\int_{\mathcal{X}\times\Theta}\psi\ d(\delta_x\times P_{\Theta|X=x})=\int_\Theta\psi(x,\theta)dP_{\Theta|X=x}(\theta)=\frac{1}{g(x)}\int_\Theta f(x|\theta)\psi(x,\theta)d\mu_\Theta(\theta)=E(\psi\mid X=x)$$

so indeed $\delta_x\times P_{\Theta|X=x}$ is the conditional probability of P given $X=x$. □

Proposition 2.4. For $A\subset\mathcal{X}$ we have that

$$P(A\times\Theta\mid\Theta=\theta)=P_\theta(A).$$

Proof. By symmetry, if we swap the roles of $\mathcal{X}$ with Θ and apply the proposition above, then we get

$$\begin{aligned}P(A\times\Theta\mid\Theta=\theta)&=\left(\int_{\mathcal{X}}f(x|\theta)d\nu(x)\right)^{-1}\left(\int_A f(x|\theta)d\nu(x)\right)\\&=P_\theta(X)^{-1}P_\theta(A)\\&=P_\theta(A).\end{aligned}$$

□

2.1. **Notation.** We use the standard notation from probability theory: if $F_i:\Omega\to\Omega_i$ are measurable and P is a measure on Ω, then

$$P(F_1\in A_1,\dots,F_n\in A_n):=P(\{\omega\in\Omega\mid F_1(\omega)\in A_1,\dots,F_n(\omega)\in A_n\})$$

and likewise for conditional measures.

Thus if P is the measure on $\mathcal{X}\times\Omega$ as constructed above, then

$$P(X\in A,\Theta\in B)=P(A\times B).$$

3. Sufficient statistics

3.1. **Basic definitions and examples.** A statistic is a measurable function $T:\mathcal{X}\to\mathcal{T}$ to some measurable space $(\mathcal{T},\mathcal{B}_\mathcal{T})$.

Definition 3.1. We say that a statistic T on $\mathcal{X}$ is *sufficient (in the Bayesian sense)* if for any prior μ_Θ on Θ we have for any Borel $B\subset\Theta$ the equality

$$P(\Theta\in B\mid X=x)=P(\Theta\in B\mid T\circ X=Tx)\quad\text{for }\mu_X\text{-almost all }x\in\mathcal{X}.$$

Example 3.2. Consider the case where $\mathcal{X} = \{0,1\}^n$ and $\Theta = [0,1]$. Suppose that P_θ is the Binomial probability distribution on $\{0,1\}^n$ where $(X_1, \ldots, X_N) \in \mathcal{X}$ is drawn i.i.d with $P(X_i = 1) = \theta$. Let us for now assume the Prior distribution μ_Θ is unknown. We know that

$$P_\theta(X = x) = \theta^{T(x)}(1-\theta)^{n-T(x)}$$

where $T(x) = \sum_{i=1}^n x_i$ is the number of successes. The posterior distribution is thus given by

$$P(\Theta \in B \mid X = x) = \frac{P(\Theta \in B, X = x)}{P(X = x)} = \frac{\int_B \theta^{T(x)}(1-\theta)^{n-T(x)} d\mu_\Theta(\theta)}{\int_\Omega \theta^{T(x)}(1-\theta)^{n-T(x)} d\mu_\Theta(\theta)}$$

for all $x \in \mathcal{X}$ that occur with positive probability (this is all of $\mathcal{X}$ except for possibly $x = (0, \ldots, 0)$ or $x = (1, \ldots, 1)$ in case θ is concentrated on 1 or 0 respectively).

We claim that $T : \mathcal{X} \to \mathbb{Z}_{\geq 0}$ given by $T(x) = \sum_{i=1}^n x_i$ is a sufficient statistic. Indeed if we fix $x_0 \in \mathcal{X} = \{0,1\}^n$ and we let $m = Tx_0$, we get that

$$\begin{aligned} P(\Theta \in B \mid T \circ X = m) &= \frac{P(\Theta \in B, T \circ X = m)}{P(T \circ X = m)} \\ &= \frac{\sum_{Tx=m} P(\Theta \in B, X = x)}{\sum_{Tx=m} P(X = x)} \\ &= \frac{\sum_{Tx=m} \int_B \theta^{T(x)}(1-\theta)^{n-T(x)} d\mu_\Theta(\theta)}{\sum_{Tx=m} \int_\Omega \theta^{T(x)}(1-\theta)^{n-T(x)} d\mu_\Theta(\theta)} \\ &= \frac{\binom{n}{m} \int_B \theta^m (1-\theta)^{n-m} d\mu_\Theta(\theta)}{\binom{n}{m} \int_\Omega \theta^m (1-\theta)^{n-m} d\mu_\Theta(\theta)} \end{aligned}$$

And so we see that this fraction cancels out to the expression for $P(\Theta \in B \mid X = x_0)$ that we already obtained.

In fact, the same argument shows that T is still a sufficient statistic if instead of assuming that P_θ is binomial we assume that P_θ is *exchangable*, meaning a distribution on $\mathcal{X} = \{0,1\}^n$ that is invariant under permutations of the co-ordinates, in this case the same calculation will instead yield

$$P(\Theta \in B \mid TX = m) = \frac{\sum_{Tx=m} \int_B P_\theta(x) d\mu_\Theta}{\sum_{Tx=m} \int_\Omega P_\theta(x) d\mu_\Theta}$$

but since we assume that $P_\Theta(x) = P_\Theta(x')$ whenever x and x' are permutations of each other, we must have that all the summands are the same and hence we get the same cancellation.

Remark 3.3. Intuitively, sufficiency of a statistic essentially means that if we want to estimate the Posterior distribution of θ given an observable statistic, we only need to look at the statistic of $T \circ X$ rather than X, which may be simpler. In other words, somebody who wishes to estimate θ (e.g., find a confidence interval) need not know the sample x but just needs to know Tx. In this example, the person wishing to estimate θ needs only to be told a single integer Tx, rather than a potentially very long sequence $x \in \{0,1\}^n$ (data compression).

The classical definition of sufficiency does not make reference to a prior distribution and is thus not Bayesian.

Definition 3.4. We say that a statistic $T : \mathcal{X} \to \mathcal{T}$ is *sufficient in the classical sense* if we have a function $r : \mathcal{B} \times \mathcal{T} \to [0,1]$ such that for all $\theta \in \Theta$ and $B \in \mathcal{B}$ (recall $\mathcal{B}$ is the Borel σ-algebra on $\mathcal{X}$) we have that

$$P_\theta(B|T = t) = r(B,t) \quad \text{for } T^*P_\theta - \text{almost all } t \in \mathcal{T}.$$

Example 3.5. Let us continue from the binomial distribution in Example 3.2. As $\mathcal{X} = \{0,1\}^n$ is a finite set, we only have to consider the case where $B \subset \mathcal{X}$ is a singleton $B = \{x_0\}$ for some $x_0 \in \mathcal{X}$. Now $P_\theta(B|T = t) = 0$ if $Tx_0 \neq t$. If $Tx_0 = t$ then we have that

$$P_\theta(B|T = t) = \frac{P_\Theta\{x_0\}}{P_\theta\{x \in \mathcal{X} \mid Tx = x_0\}} = \frac{1}{\binom{n}{t}}$$

where the last equality follows from the fact that all $x \in \mathcal{X}$ with $Tx = t$ have the same probability in a binomial distribution. In fact, the same argument applies to the case where the P_θ are all exchangable.

Thus we may take

$$r(\{x_0\}, t) = \mathbb{1}(Tx_0 = t)\binom{n}{t}^{-1}$$

and extend r to $\mathcal{B} \times \mathbb{Z}$ by additivity. Thus this parametric family is indeed sufficient in the classical sense.

3.2. **Equivalence of definitions.**

Lemma 3.6. Suppose $T : \mathcal{X} \to \mathcal{T}$ is a statistic and μ_Θ is a prior on Θ. Suppose also that $(\theta, t) \mapsto P_\theta(A \mid T = t)$ is Borel (i.e., there is a version of conditional probabilities so that it is Borel). Then for $A \subset \mathcal{X}$ we have that

$$P(X \in A \mid TX = Tx, \Theta = \theta) = P_\theta(A \mid T = Tx) \quad \text{for } P\text{-almost all } (x, \theta) \in \mathcal{X} \times \Theta.$$

Proof. We must show that for $B \subset \Theta, C \subset \mathcal{T}$ we have that

$$\int_{T^{-1}C \times B} P(A \times \Theta \mid Tx = t, \Theta = \theta) dP(x, \theta) = \int_{T^{-1}C \times B} P_\theta(A \mid T = Tx) dP(x, \theta).$$

The law of total probability tells us that the left hand side is equal to

$$P((A \times \Theta) \cap (T^{-1}C \times B)) = P(X \in A \cap T^{-1}C, \Theta \in B).$$

The right hand side is

$$\begin{aligned}
\int_B \left(\int_{T^{-1}C} P_\theta(A|T = Tx) dP_\theta(x) \right) d\mu_\Theta(\theta) &= \int_B P_\theta(A \cap T^{-1}C) d\mu_\Theta(\theta) \\
&= \int_B \int_{\mathcal{X}} \mathbb{1}_{A \cap T^{-1}C}(x) dP_\theta(x) d\mu_\Theta(\theta) \\
&= P(X \in A \cap T^{-1}C, \Theta \in B)
\end{aligned}$$

□

Theorem 3.7. Suppose that $T : \mathcal{X} \to \mathcal{T}$ is sufficient in the classical sense. Then T is also sufficient in the Bayesian sense (for any prior μ_Θ on Θ). Moreover for $A \subset \mathcal{X}$ we have

$$P(X \in A \mid TX = Tx, \Theta = \theta) = P(X \in A \mid TX = Tx) \quad \text{for } P\text{-almost all } (x, \theta) \in \mathcal{X} \times \Theta.$$

Proof. We first show the *Moreover* claim. By the previous lemma we have that the left hand side is equal to

$$P_\theta(A \mid T = Tx) = r(A, Tx)$$

for some measurable $t \mapsto r(A, t)$. So now it remains to show that $r(A, Tx) = P(X \in A \mid TX = Tx)$ for almost P_X-almost all x. Thus we must show that for $C \subset \mathcal{T}$ we have that

$$\int_{T^{-1}C} r(A, Tx) dP_X(x) = P_X(A \cap T^{-1}C). \tag{1}$$

In the proof of the previous Lemma (take $B = \Theta$ there) we have shown that

$$\int_{T^{-1}C\times\Theta} P_\theta(A|T = Tx)dP(x, \theta) = P_X(A \cap T^{-1}C).$$

This completes the proof of (1) since $r(A, Tx) = P_\theta(T = Tx)$ holds for almost all P-almost all (x, θ). Now it remains to show that T is sufficient in the Bayesian sense. By this just proved Moreover claim, it follows from Lemma D.4 that X and Θ are conditionally independent given TX. Using Lemma D.4 again we have that

$$P(\Theta \in B \mid TX = Tx, X = x) = P(\Theta \in B \mid TX = Tx) \quad \text{for } P_X\text{-almost all } x.$$

But the left hand side is clearly $P(\Theta \in B \mid X = x)$. Thus T is indeed sufficient in the Bayesian sense. □

Lemma 3.8. A statistic $T : \mathcal{X} \to \mathcal{T}$ is sufficient in the Bayesian sense if and only if for all prior distributions μ_Θ on Θ and measurable $B \subset \mathcal{X}$ we have that

$$x \mapsto P(\Theta \in B \mid x \in X)$$

is almost surely measurely $T^{-1}\mathcal{B}_\mathcal{T}$-measurable (i.e., almost surely equal to a function of T).

Proof. We prove the non-obvious direction only: Thus suppose that for each measurable $B \subset X$ we have that $x \mapsto P(\Theta \in B \mid x \in X) = h_B(Tx)$ where $h_B : \mathcal{T} \to \mathbb{R}$ is Borel. We first use apply the law of total probability to the map $T \circ X : \mathcal{X} \times \Theta \to \mathcal{T}$ as follows. For $A \subset \mathcal{T}$ Borel we have that

$$P(\Theta \in B, T \circ X \in A) = \int_A P(\Theta \in B|T \circ X = t)dP_\mathcal{T}(t).$$

Now we apply the law of total probability to the map $X : \mathcal{X} \times \Theta \to \mathcal{X}$ to obtain that

$$\begin{aligned}
P(\Theta \in B, T \circ X \in A) &= P(\Theta \in B, X \in T^{-1}A) \\
&= \int_{T^{-1}A} P(\Theta \in B|X = x)dP_X(x) \\
&= \int_{T^{-1}A} h_B(Tx)dP_X(x) \\
&= \int_A h_B(t)dP_\mathcal{T}(t).
\end{aligned}$$

Thus we must have that $P(\Theta \in B|T \circ X = t) = h_B(t)$ holds for almost all t. Thus

$$P(\Theta \in B \mid T \circ X = Tx) = h_B(Tx) = P(\Theta \in B \mid X = x)$$

holds for almost all $x \in X$. □

Lemma 3.9. Suppose that ν is some σ-finite measure on $\mathcal{X}$ such that $P_\theta \ll \nu$ for all $\theta \in \Theta$. If $T : \mathcal{X} \to \mathcal{T}$ is sufficient in the Bayesian sense then there exists a probability measure ν^* on $\mathcal{X}$ such that $P_\theta \ll \nu^*$ for all $\theta \in \Theta$, $\nu^* \ll \nu$ and for each $\theta \in \Theta$ there is a Borel

$$h_\theta : \mathcal{T} \to [0, \infty)$$

such that

$$\frac{dP_\theta}{d\nu^*}(x) = h_\theta(Tx) \quad \text{for } \nu^*\text{-almost all } x.$$

Proof. By Lemma E.2 there exists a countable or finite sequence of distinct $\theta_1, \theta_2, \dots$ and $0 < c_1, c_2, \dots \le 1$ with $\sum_i c_i = 1$ such that $\nu^* := c_i P_{\theta_i}$ satisfies that $P_\theta \ll \nu^* \ll \nu$ for all $\theta \in \Theta$. Now fix any $\theta_0 \in \Theta$. First suppose $\theta_0 \neq \theta_i$ for all $i > 0$. Define now a prior μ on Θ where $\mu(\{\theta_0\}) = 1/2$ and $\mu(\{\theta_i\}) = c_i/2$, i.e., μ is discrete. Now we have

$$\begin{aligned} P(\Theta = \theta_0 \mid X = x) &= \frac{\int_{\{\theta_0\}} f(x|\theta) d\mu(\theta)}{\int_\Theta f(x|\theta) d\mu(\theta)} \\ &= \frac{\frac{1}{2} f(x|\theta_0)}{\frac{1}{2} f(x|\theta_0) + \frac{1}{2}\sum_{i>0} c_i f(x|\theta_i)}. \end{aligned}$$

Observe that the pushfoward measure on $\mathcal{X}$ is $P_X = \frac{1}{2}P_{\theta_0} + \frac{1}{2}\nu^*$. We know by Lemma 2.2 that for P_X-almost all $x \in X$ the denominator is non-zero. Now observe that

$$\frac{d\nu^*}{d\nu}(x) = \sum_{i>0} c_i f(x|\theta_i)$$

and as $\nu^* \ll \nu$ we have that this is ν^* almost surely non-zero and thus P_X-almost surely non-zero as $P_X \ll \nu^*$ as shown above. So we can divide by this quantity to get that

$$\frac{dP_{\theta_0}}{d\nu^*(x)} = f(x|\theta_0)\left(\sum_{i>0} c_i f(x|\theta_i)\right)^{-1} \quad \text{for } P_X\text{-almost all } x \in \mathcal{X}. \tag{2}$$

Thus we have that

$$P(\Theta = \theta_0 \mid X = x) = \frac{dP_\theta/d\nu^*(x)}{dP_\theta/d\nu^*(x) + 1}$$

holds P_X-almost surely. However, P_X-almost surely we have that the right hand side is $\mathcal{T}$-measurable. Thus ν^*-almost surely it also is. This comlpetes the proof in the case $\theta_0 \neq \theta_i$ for all $i > 0$. If $\theta_0 = \theta_i$ for some $i > 0$ we instead define the prior

$$\mu = \sum_{i>0} c_i P_{\theta_i}.$$

Without loss of generality, say $\theta_0 = \theta_1$. Thus

$$P(\Theta = \theta_1 \mid X = x) = \frac{c_1 f(x|\theta_1)}{\sum_{i>0} c_i f(x|\theta_i)} = c_1 \frac{dP_{\theta_1}}{d\nu^*}(x).$$

Thus again we have that this is P_X-almost equal to a $\mathcal{T}$-measurable function (and thus ν^* almost equal to such).

$\square$

Theorem 3.10. Assume the nice condition of Definition 1.5 . If $T : \mathcal{X} \to \mathcal{T}$ is sufficient in the Bayesian sense, then it is also sufficient in the classical sense.

Proof. Let ν^* be the measure from the previous Lemma and recall that

$$\frac{dP_\theta}{d\nu^*}(x) = h_\theta(Tx).$$

We will now show that for all $\theta \in \Theta$ and $A \subset X$ we have that

$$P_\theta(A \mid T = t) = \nu^*(A \mid T = t),$$

which will complete the proof. Observe that for $C \subset \mathcal{T}$ we have that

$$
\begin{aligned}
\int_C \nu^*(A \mid T=t)d(T^*P_\theta)(t) &= \int_{T^{-1}C} \nu^*(A \mid T=Tx)dP_\theta(x) \\
&= \int_{T^{-1}C} \nu^*(A \mid T=Tx)h_\theta(Tx)d\nu^*(x) \\
&= \int_{\mathcal{X}} \mathbb{E}(\mathbb{1}_A \mid T=Tx)\mathbb{1}_{T^{-1}C}(x)h_\theta(Tx)d\nu^*(x) \\
&= \int_{\mathcal{X}} \mathbb{1}_A(x)\mathbb{1}_{T^{-1}C}(x)h_\theta(Tx)d\nu^*(x) \\
&= \int_{\mathcal{X}} \mathbb{1}_A(x)\mathbb{1}_{T^{-1}C}(x)dP_\theta(x) \qquad = P_\theta(A\cap T^{-1}C)
\end{aligned}
$$

which completes the proof by the law of total probability. □

By observing the proofs so far, we see that to check sufficiency in the Bayesian sense it is enough to check only those prior distributions that are discrete (i.e., those μ_Θ on Θ where $\mu(C)=1$ for some at most countable subset $C\subset\Theta$).

Corollary 3.11. Assuming the nice condition of Definition 1.5 . We have that $\mathcal{T}:\mathcal{X}\to\mathcal{T}$ is sufficient (in either classical of Bayesian sense) if and only if the Bayesian sufficiency condition is satisfied for all discrete prior distributions μ_Θ on Θ.

Proof. Observe that the measures μ used in the proof of Lemma 3.9 are all discrete. Once ν^* is constructed, sufficiency follows as seen in the proof of the previous Theorem. □

3.3. **Fisher-Neyman factorization.** There is a convenient characterization of sufficiency.

Theorem 3.12. Assume the nice condition of Definition 1.5. Then a statistic $\mathcal{T}:\mathcal{X}\to\mathcal{T}$ is sufficient if and only if the following holds: There is Borel $g:\mathcal{X}\to\mathbb{R}_{\geq 0}$ such that for all $\theta\in\Theta$ we have a Borel $h_\theta:\mathcal{T}\to\mathcal{R}_{\geq 0}$ such that

$$f(x|\theta)=h_\theta(Tx)g(x) \quad \text{ for } P_\theta \text{ almost all } x\in X$$

Proof. First suppose this condition holds. Thus by Corollary 3.11 we only need to check Bayesian sufficiency for any discrete prior μ on Θ. Say μ is supported on $C=\{\theta_1,\theta_2,\ldots\}\subset\Theta$. For $C'\subset C$ we have by Bayes' formula that

$$P(\Theta\in C\mid X=x)=\frac{\int_{C'}f(x|\theta)d\mu(\theta)}{\int_\Theta f(x|\theta)d\mu(\theta)}=\frac{\sum_{\theta\in C'}h_\theta(Tx)g(x)\mu(\{\theta\})}{\sum_{\theta\in C}h_\theta(Tx)g(x)\mu(\{\theta\})}=\frac{\sum_{\theta\in C'}h_\theta(Tx)\mu(\{\theta\})}{\sum_{\theta\in C}h_\theta(Tx)\mu(\{\theta\})}.$$

Observe now that this is $\mathcal{T}$-measurable since the sums are countable, thus Bayesian sufficiency is established. (**Remark:** this is the reason we restrict without loss of generality to μ discrete, as otherwise it seems one cannot show measurability in the integrand unless we also assume that $(\theta,t)\mapsto h_\theta(t)$ is Borel, which we do not want to as we have not been able to verify this in Lemma 3.9).

Conversely, suppose that $\mathcal{T}$ is Bayesian. Thus by Lemma 3.9) we have

$$f(x|\theta)=\frac{dP_\theta}{d\nu}(x)=\frac{dP_\theta}{d\nu^*}(x)\frac{d\nu^*}{d\nu}(x)=h_\theta(Tx)g(x)$$

for ν^*-almost all $x\in X$. As $P_\theta\ll\nu^*$ we have it also holds for P_θ-almost all $x\in X$. □

Example 3.13. In Example 3.2 we have that

$$f(x|\theta) = P_\theta(X = x) = \theta^{T(x)}(1-\theta)^{n-T(x)}$$

so can just take

$$h_\theta(t) = \theta^t(1-\theta)^{n-t}$$

and $g(x) = 1$. Here $\nu(\{x\}) = 1$ is the counting measure. If instead we had a different measure $\nu(\{x\}) = q(x) > 0$ then we would have

$$f(x|\theta) = P_\theta(X = x)q(x)^{-1} = h_\theta(Tx)q(x)^{-1}$$

so then $g(x) = q(x)^{-1}$. This shows that $g(x)$ is just a change of density.

Appendix A. Review of Probability

Appendix B. Borel spaces

Lemma B.1. Let X be a second-countable metric space. Then there exists an embedding $\iota : X \hookrightarrow Y$ where Y is compact and ι is a homeomorphism onto its image $\iota(X)$.

Proof. Let $u_1, u_2, \dots$ be a countable dense subset of X. Suppose that $d(x, x') \leq 1$ for all x, x'. Define

$$\iota : X \to [0,1]^{\mathbb{N}}$$

to be

$$\iota(x) = (d(x, u_i))_{i=1}^{\infty}$$

where $Y = [0,1]^{\mathbb{N}}$ has the product topology, thus is a compact metric space. Continuity and injectivity of ι is clear. Now suppose that $x_1, x_2, \dots \in X$ and $x \in X$ are such that $\iota(x_i) \to \iota(x)$ as $i \to \infty$. Fix $\epsilon > 0$ and choose u_j so that $d(x, u_j) < \epsilon$. Now for all sufficiently large i we have that $|d(x_i, u_j) - d(x, u_j)| < \epsilon$. This implies that $d(x_i, u_j) < 2\epsilon$. Now

$$d(x, x_i) \leq d(x, u_j) + d(u_j, x_i) \leq \epsilon + 2\epsilon = 3\epsilon.$$

Thus ι^{-1} is continuous on $\iota(X)$. □

Lemma B.2. Let X be a complete metric space, with metric d_X, and suppose that $Y \subset X$ is a subset of X such that the subspace topology on Y is induced by a complete metric d_Y on Y. Then Y is a countable intersection of open subsets of X, and is thus a Borel subset of Y.

Proof. We first show that for each $\epsilon > 0$ and $y \in Y$ there exists an open set $V_{y,\epsilon}$ of X such that $y \in V$ and $\mathrm{diam}_{d_X}(V_{y,\epsilon}) \leq \epsilon$ and $\mathrm{diam}_{d_Y}(V_{y,\epsilon} \cap Y) \leq \epsilon$. To see this, observe since d_X and d_Y induce the same topology on Y, there is an open set $\tilde{V}$ of X such that $\tilde{V} \cap Y$ is the ball of radius $\epsilon/2$ around y in the d_Y metric. Hence we can take

$$V_{y,\epsilon} = \tilde{V} \cap B_{d_X}(y, \epsilon/2).$$

Now let $U_n = \bigcup_{y \in Y} V_{y,1/n}$ (Remark: to avoid axiom of choice, just take the union of y and all the different sets of the form $\tilde{V}$). Observe now that $Y \subset U_n$ for all n. We claim that

$$Y = \bigcap_{n=1}^{\infty} U_n.$$

Thus suppose that $x \in U_n$ for all $n \geq 1$. We must show $x \in Y$. We will show that, by induction on n, that there exists a sequence of open subsets $W_1 \supset W_2 \supset \dots$ of X and $y_n \in Y \cap W_n$ such that

$\mathrm{diam}_{d_X}(W_n) \leq \frac{1}{n}, \mathrm{diam}_{d_Y}(W_n \cup Y) \leq \frac{1}{n}$, $x \in W_n$. We can take $W_1 = V_{y_1,1}$ for some $y_1 \in Y$ since $x \in U_1$. Now suppose W_n has been constructed. Thus there exists $\epsilon > 0$ so that $B_{d_X}(x, \epsilon) \subset W_n$ as W_n is open and $x \in W_n$. Now take $N > n$ such that $1/N < \epsilon$. Now since $x \in U_N$ there exists $y_{n+1} \in Y$ and $W_{n+1} \subset X$ open so that $y_{n+1}, x \in W_{n+1}$ and $\mathrm{diam}_{d_X}(W_{n+1}) \leq 1/N$, $\mathrm{diam}_{d_Y}(W_{n+1} \cap Y) \leq 1/N$. Now observe that $W_{n+1} \subset B_{d_X}(x, \epsilon) \subset W_n$ since if $x' \in W_{n+1}$ then

$$d_X(x, x') \leq \mathrm{diam}_{d_X}(W_{n+1}) \leq 1/N < \epsilon.$$

Thus in summary $y_{n+1} \in W_{n+1}$ is non-empty as desired and $W_{n+1} \subset W_n$ and

$$\mathrm{diam}_{d_Y}(W_{n+1} \cap Y) \leq \frac{1}{N} \leq \frac{1}{n+1}.$$

It now follows that $y_n \to x$ in d_X metric and that $y_1, y_2, \ldots$ is a Cauchy sequence in Y. Thus by completeness, we have that $y_n \to y \in Y$ for some $y \in Y$. However

$$d_X(x, y) \leq d_X(x, y_n) + d_X(y_n, y) \to 0$$

as $n \to \infty$ since $d_Y(y, y_n) \to 0$ and d_X and d_Y induce the same topology. Thus $x = y \in Y$ as required. $\square$

Corollary B.3. Let X be a second countable complete metric space. Then X is homeomorphic to a Borel subset of a compact metric space. In fact, we can take this Borel subset to be a countable intersection of open subsets.

Definition B.4. A *standard Borel space* is a measurable space $(X, \mathcal{B})$ where X is a second-countable complete metric space and $\mathcal{B}$ is the Borel σ-algebra.

Appendix C. Conditional expectation and probability

C.1. Conditional expectation.

Theorem C.1. Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subset \mathcal{B}$ be a sub-σ-algebra of $\mathcal{B}$. Then for $f \in L^1(X, \mathcal{B}, \mu)$ there exists a unique $g \in L^1(X, \mathcal{A}, \mu)$ such that

$$\int_A f d\mu = \int_A g d\mu \quad \text{for all } A \in \mathcal{A}.$$

Definition C.2. We say that the $g \in L^1(X, \mathcal{B}, \mu)$ the *conditional expectation of f w.r.t.* $\mathcal{A}$, and denote it by $g = E(f|\mathcal{A})$.

Proof. If $f \in L^2(X, \mathcal{B}, \mu)$ then we define $E(f|\mathcal{A})$ to be the orthogonal projection of f onto the closed subspace $L^2(X, \mathcal{A}, \mu)$. Indeed, for $A \in \mathcal{A}$ we have that $\mathbb{1}_A \in L^2(X, \mathcal{A}, \mu)$ and so

$$\int_A f d\mu = \langle f, \mathbb{1}_A \rangle = \langle E(f|\mathcal{A}), \mathbb{1}_A \rangle = \int_A E(f|\mathcal{A}) d\mu$$

as required. We will now extend $E(\cdot|\mathcal{A})$ to L^1. By density of L^2 in L^1 it is enough to show that $E(\cdot|\mathcal{A})$ is a bounded operator w.r.t. to the L^1 norm on L^2. For $f \in L^2(X, \mathcal{B}, \mu)$ let $\psi \in L^\infty(X, \mathcal{A}, \mu)$ be such that $|E(f|\mathcal{A})| = \psi E(f|\mathcal{A})$ and $\|\psi\|_\infty \leq 1$. Thus

$$\|E(f|\mathcal{A})\|_1 = \int_X \psi E(f|\mathcal{A}) d\mu = \langle E(f|\mathcal{A}), \overline{\psi} \rangle = \langle f, \overline{\psi} \rangle = \int_X \psi f d\mu \leq \|f\|_1.$$

Thus $E(\cdot|\mathcal{A})$ has L^1 operator norm at most 1 and thus can be extended to $L^1(X, \mathcal{B}, \mu) \to L^1(X, \mathcal{A}, \mu)$.

For uniqueness, suppose that $g_1, g_2 \in L^1(X, \mathcal{A}, \mu)$ are real valued functions such that

$$\int_A g_1 d\mu = \int_A g_2 d\mu \quad \text{for all } A \in \mathcal{A}.$$

Let

$$A_\epsilon = \{x \in X \mid g_1(x) \leq g_2(x) - \epsilon\}$$

and observe that $A_\epsilon \in \mathcal{A}$. Thus

$$\int_A g_2 = \int_A g_1 d\mu \leq \int_A g_2 d\mu - \epsilon\mu(A)$$

which implies that $\mu(A_\epsilon) = 0$. Thus $g_1 \geq g_2$ almost everywhere. By symmetry, $g_1 = g_2$ almost everywhere. For g_1, g_2 complex valued uniqueness also follows by equating real and imaginary parts. $\square$

Definition C.3. Let $T : X \to Y$ be a measurable map between measure spaces $(X, \mathcal{B}_X)$ and $(Y, \mathcal{B}_Y)$. We define $T^{-1}\mathcal{B}_Y = \{T^{-1}B \mid B \in \mathcal{B}_Y\}$.

Proposition C.4. Suppose that $(X, \mathcal{B}_X, \mu_X)$ is a probability space, $(Y, \mathcal{B}_Y)$ is a measurable space $T : X \to Y$ is measurable. Then any function $f : X \to \mathbb{R}$ that is $T^{-1}\mathcal{B}_Y$ measurable (that is, $f^{-1}B \in T^{-1}\mathcal{B}_Y$ for all Borel $B \subset \mathbb{R}$) can be written as $f = g \circ T$ for some measurable $g : Y \to \mathbb{R}$.

In other words, if we let $\mu_Y = T^*\mu_X$ be the push-forward measure then we have an isometric embedding

$$L^p(Y, \mathcal{B}_Y, \mu_Y) \hookrightarrow L^p(X, \mathcal{B}_X, \mu_X)$$

given by $g \mapsto g \circ T$ whose image is exactly $L^p(\mathcal{X}, T^{-1}\mathcal{B}_Y, \mu_X)$.

Proof. If $f = \mathbb{1}_A$ for some $A \in T^{-1}\mathcal{B}_Y$ then $A = T^{-1}B$ for some $B \in \mathcal{B}_Y$ and so we can take $g = \mathbb{1}_B$. For $f \geq 0$ note that we can write

$$f = \sum_{i=1}^{\infty} c_i \mathbb{1}_{A_i}$$

for some $c_i \geq 0$ and $A_i \in T^{-1}\mathcal{B}_y$ where the limit is pointwise and so $\mathbb{1}_{A_i} = \mathbb{1}_{B_i} \circ T$ for some $B_i \in \mathcal{B}_Y$. Thus $f = g \circ T$ where

$$g(y) = \sum_{i=1}^{\infty} c_i \mathbb{1}_{B_i}(y)$$

converges pointwise as it is monotonic $c_i > 0$. For arbitrary measurable f, note that we can write it as a linear combination of non-negative functions.

Now note that for $g \geq 0$ we have that

$$\int_X g \circ T d\mu_X = \int_Y g d\mu_Y$$

which can be shown as follows. Writing

$$g(y) = \sum_{i=1}^{\infty} c_i \mathbb{1}_{B_i}(y)$$

and using the monotone convergence theorem we have that

$$\int_X g(Tx) d\mu_X(x) = \sum_{i=1}^{\infty} c_i \int_X \mathbb{1}_{B_i}(Tx) d\mu_X(x) = \sum_{i=1}^{\infty} c_i \mu_X(T^{-1}B_i) = \sum_{i=1}^{\infty} c_i \mu_Y(B_i) = \int_Y g(y) d\mu_Y(y).$$

This shows that for $1 \leq p < \infty$ we have that

$$\|g \circ T\|_p^p = \int_X |g|^p(Tx)\mu_X(x) = \int_Y |g|^p(y) d\mu_Y(y) = \|g\|_p^p$$

and so indeed the embedding is an isometry. $\square$

This proposition motivates us to define conditional expectation with respect to a random variable (or factor map).

Definition C.5. Let $(X, \mathcal{B}, \mu)$ be a probability space and suppose that $T : X \to Y$ is measurable where $(Y, \mathcal{B}_Y)$ is a measure space (i.e., a random variable). Then for $f \in L^p(\mathcal{X}, \mathcal{B}, \mu)$ (for $p = 1, 2, \infty$) we define a function on $Y \to \mathbb{C}$ denoted $y \mapsto E(f|T = y)$ that is characterized for μ-almost all x by

$$E(f|\mathcal{T}^{-1}\mathcal{B}_Y)(x) = E(f|T = y) \quad \text{where } y = Tx.$$

In other words, the function $E(f|\mathcal{T}^{-1}\mathcal{B}_Y) \in L^p(\mathcal{X}, T^{-1}\mathcal{B}_Y, \mu)$ is identified with a function in $L^p(Y, \mathcal{B}_Y, T^*\mu)$ using the isometry in the proposition above and this function at $y \in Y$ is denoted $E(f|T = y)$. Note that $E(f|T = y)$ is uniquely defined for $T^*\mu_X$-almost all $y \in Y$.

Note that $E(f|T = y)$ can be characterized by the property that

$$\int_A E(f|T = y) d\mu_Y(y) = \int_{T^{-1}A} f(x) d\mu_X(x) \quad \text{for all } A \in \mathcal{B}_Y.$$

C.2. **Conditional Probability.**

Theorem C.6 (Conditional probability)**.** Let X be a second-countable complete metric space with Borel σ-algebra $\mathcal{B}$ and suppose $\mathcal{A} \subset \mathcal{B}$ is a sub σ-algebra. Let μ be a probability measure on $(X, \mathcal{B})$. Then there exists $X' \in \mathcal{A}$ with $\mu(X') = 1$ and a family of probability measures $\mu_x^{\mathcal{A}}$ for $x \in X'$ such that the following holds.

- For $\mathcal{B}$-measurable functions $f : X \to \mathbb{C}$ such that $\|f\|_\infty < \infty$, we have that the mapping $X' \to \mathbb{C}$ given by
$$x \mapsto \int_X f(u) d\mu_x^{\mathcal{A}}(u)$$
is $\mathcal{A}$-measurable. The same is true if we replace $\|f\|_\infty < \infty$ with $f(x) \in \mathbb{R}_{\geq 0}$ for all $x \in X$.
- For $f : X \to \mathbb{C}$ with $\|f\|_1 < \infty$ we have that[1]
$$E(f|\mathcal{A})(x) = \int_X f(u) d\mu_x^{\mathcal{A}}(u) \quad \text{for almost all } x \in X.$$

Definition C.7. We call such a family of measures $\mu_x^{\mathcal{A}}$ *conditional probabilities of* μ *w.r.t* $\mathcal{A}$.

Proof. Using Corollary B.3 we may assume X is compact. Thus there exists a countable dense $\mathbb{Q}$-vector subspace V of $C(X)$ such that $1 \in V$. Now for each $f \in V$ choose a geniune function (not an equivalence class of functions) $g_f : X \to \mathbb{C}$ such that

$$g_f(x) = E(f|\mathcal{A})(x)$$

and

$$\|g_f\|_\infty \leq \|f\|_\infty.$$

Now for any $\alpha, \beta \in \mathbb{Q}$ and $f_1, f_2 \in V$ we have that

$$g_{\alpha f_1 + \beta f_2}(x) = \alpha g_{f_1}(x) + \beta g_{f_2}(x) \quad \text{for almost all } x \in X.$$

We also have $g_1(x) = 1$ and $g_f(x) \geq 0$ for positive $f \in V$ (i.e, $f \geq 0$) for almost all $x \in X$. Thus we may choose $X' \subset X$ to be a set with $\mu(X') = 1$ on which all of thse properties hold (as there are only countably many equations). Thus for each $x \in X'$ we have a linear function $\Lambda_x' : V \to \mathbb{C}$ given by

$$\Lambda_x'(f) = g_f(x).$$

[1] The right hand side may be undefined on a null set of x.

Note that Λ'_x is uniformly continuous (since $\|g_f\|_\infty \leq \|f\|_\infty$) and thus it extends to a linear functional $\Lambda_x : C(X) = \overline{V} \to \mathbb{C}$ which also satisfies that

$$|\Lambda_x(f)| \leq \|f\|_\infty.$$

We wish to apply the Riesz representation theorem, thus it remains to show that Λ_x is a positive linear functional. Now suppose $f \in C(X)$ is positive. Then there exists $f_i \in V$ such that $f_i \to f$ uniformly. Now fix a positive integer q and let $\epsilon = 1/q$. Then $\Lambda_x(\epsilon) = \epsilon$ since $g_1 = 1$. For i sufficiently large we have that we have $\|f_i - f\|_\infty < \epsilon$ and so $f_i + \epsilon \geq 0$ as f is positive. As $f_i + \epsilon \in V$ we have that

$$\Lambda_x(f_i) + \epsilon = g_{f_i+\epsilon}(x) \geq 0.$$

Thus

$$\Lambda_x(f) \geq \Lambda_x(f_i) - \Lambda_x(|f_i - f|) \geq \Lambda_x(f_i) - \epsilon > -2\epsilon.$$

This shows that $\Lambda_x(f) \geq 0$ and thust Λ_x is a positive linear continuous functional on $C(X)$ with $\Lambda_x(1) = 1$. Thus there exists a measure μ_x such that

$$\Lambda_x(f) = \int_X f(u) d\mu_x(u) \quad \text{for all } f \in C(X), x \in X'.$$

In particular this means that

$$E(f|\mathcal{A})(x) = \int_X f(u) d\mu_x(u) \quad \text{for all } f \in V.$$

This means that

(3)
$$x \mapsto \int_{X'} f(u) d\mu_x(u) \text{ is } \mathcal{A}\text{-measurable on } X'$$

and

(4)
$$\int_A \int_{X'} f(u) d\mu_x(u) d\mu(x) = \int_A f(x) d\mu(x) \qquad \text{for all } A \in \mathcal{A}$$

holds for all $f \in V$. By density of V in $C(X)$, this also holds for all $f \in C(X)$ and the integrand is indeed $\mathcal{A}$ measurable (it is a pointwise limit of $\mathcal{A}$ measurable functions). Since the indicator function of an open set is a monotonic limit of continuous functions, we have that (3) and (4) holds for $f = \mathbb{1}_U$ for all $U \subset X$ open. It also holds for closed sets by linearity. Finally, observe that it also holds for an intersection $U \cap C$ where U is open and C is closed since such a set is a countable nested intersection of open sets.

Now let

$$\mathcal{R} = \{\cup_{i=1}^n U_i \cap C_i \mid U_1, \ldots, U_n \text{ are open and } C_1, \ldots, C_n \text{ are closed.}\}$$

be the set of all finite unions of sets that can be written as an intersection of an open set and a closed set. We claim that any element in $\mathcal{R}$ can be written as a finite disjoint union of an intersection of an open set with a closed set. Indeed, any such finite union belongs to a finite σ-algebra generated by such sets but any atom of this σ-algebra must take this form. It now follows that (3) and (4) holds for $f = \mathbb{1}_R$ for all $R \in \mathcal{R}$. By the monotone convergence theorem, these also hold on the monotone class generated by $\mathcal{R}$. Thus by the monotone class lemma, it holds on all Borel sets since $\mathcal{R}$ is an algebra. Any measurable $f : X \to \mathbb{R}_{\geq 0}$ is a monotonic limit of simple functions and so it holds for such f too, which completes the proof for such f. The theorem now follows for general $f : X \to \mathbb{C}$ with $\|f\|_1$ as desired by linearity. $\square$

We now show that the conditional probabilities are almost surely unique and provide some characterizations that are practial to use. In the following proposition, we fix $X, \mathcal{A}, \mu$ as as in Theorem C.6.

Proposition C.8. If $\mu_x^{\mathcal{A}}$ and $\nu_x^{\mathcal{A}}$ are two conditional probabilities, for some $x \in X' \in \mathcal{A}$ of full measure, then $\mu_x^{\mathcal{A}} = \nu_x^{\mathcal{A}}$ for μ-almost all $x \in X$.

Proof. We for each $f \in C(X)$ we have that

$$\int_X f(u) d\mu_x^{\mathcal{A}}(u) = E(f|\mathcal{A})(x) = \int_X f(u) d\nu_x^{\mathcal{A}}(u)$$

holds for all $x \in X_f \in \mathcal{A}$ with $\mu(X_f) = 1$. Applying this to a countable dense set of $f \in C(X)$ we can find a single set $X' \in \mathcal{A}$ with $\mu(X'') = 1$ so that this identity holds for all continuous functions $f \in C(X)$. It follows by uniqueness in Riesz-representation theorem that $\mu_x^{\mathcal{A}} = \nu_x^{\mathcal{A}}$ for all $x \in X''$.

□

C.3. **Conditional probability with respect to a random variable.** In this subsection, $\mathcal{A}$ is a countable generated σ-algebra of $\mathcal{B}$ where $(X, \mathcal{B})$ is standard Borel. So $\mathcal{A} = \sigma(\{A_1, A_2, \ldots, \})$ where $A_1, A_2, \ldots \in \mathcal{A}$. For each $x \in X$ we can define the atom

$$[x]_{\mathcal{A}} = \bigcap_{i, x \in A_i} A_i \cap \bigcap_{j, x \notin A_j} (X \setminus A_j)$$

containing x.

Proposition C.9. For $x \in X$ we have that $[x]$ is the smallest subset of $\mathcal{A}$ containing x.

Proof. We must show that for each $A \in \mathcal{A}$ we have either $[x] \subset \mathcal{A}$ or $[x] \cap \mathcal{A} = \emptyset$. This is true for each A_i and so one must show that the sets in $\mathcal{A}$ satisfying this property forms a σ-algebra. □

Proposition C.10. There is a set $X'' \subset X$ with $X'' \in \mathcal{A}$ and $\mu(X'') = 1$ such that for any $x_1, x_2 \in X''$ we have that $\mu_{x_1}^{\mathcal{A}} = \mu_{x_2}^{\mathcal{A}}$ if and only if $[x_1] = [x_2]$. Moreover, $\mu_x^{\mathcal{A}}([x]) = 1$ for $x \in X''$.

Proof. Let $\mu_x = \mu_x^{\mathcal{A}}$ for $x \in X'$ be a set of conditional probabilities, where $X' \in \mathcal{A}$ and $\mu(X') = 1$. For almost all $x \in X'$ we have that

$$\mu_x(A_i) = \int \mathbb{1}_{A_i} d\mu_x = E(\mathbb{1}_{A_i}|\mathcal{A})(x) = \mathbb{1}_{A_i}(x).$$

Say this occurs on $X'' \subset X'$ with $\mu(X'') = 1$. Note that since all terms are $\mathcal{A}$-measurable, we have $X'' \in \mathcal{A}$. Now let $x_1, x_2 \in X''$. If $\mu_{x_1} = \mu_{x_2}$ then $\mathbb{1}_{A_i}(x_1) = \mathbb{1}_{A_i}(x_2)$ for all i and thus $[x_1] = [x_2]$. Conversely, suppose that $[x_1] = [x_2]$. Thus any $\mathcal{A}$-measurable function must have equal values on x_1 and x_2. But for any $f \in C(X)$ we have that

$$x \mapsto \int f d\mu_x$$

is $\mathcal{A}$-measurable. Thus

$$\int f d\mu_{x_1} = \int f d\mu_{x_2}$$

holds for all $f \in C(X)$ and thus $\mu_{x_1} = \mu_{x_2}$. □

Theorem C.11 (Law of total probability)**.** Suppose that $T : X \to Y$ is a measurable map between standard Borel spaces and let $\mathcal{B}_T := T^{-1}\mathcal{B}_Y$ be the induced σ-algebra. There exists Borel $Y' \subset Y$ and probability measures μ_y on X, for each $y \in Y'$, such that the following hold.

- $\mu(T^{-1}Y') = 1$
- For all $y \in Y'$, the measure μ_y is supported on $T^{-1}(\{y\})$.

- For all $x \in T^{-1}(Y')$ we have that $\mu_{Tx} = \mu_x^{\mathcal{B}_T}$. That is the family μ_{Tx} for $x \in T^{-1}(Y')$ is a family of conditional probability measures for the σ-algebra $\mathcal{B}_T$.

Moreover, for bounded Borel $f : X \to \mathbb{C}$ and $B \subset Y$ Borel we have that

$$\int_{T^{-1}B} f(x)d\mu(x) = \int_B \left(\int_X f(u)d\mu_y(u) \right) d(T^*\mu)(y).$$

Proof. The induced σ-algebra $\mathcal{B}_T := T^{-1}\mathcal{B}_Y$ is countably generated and its atoms are precisely the fibers

$$T^{-1}(\{y\}) \quad \text{for } y \in Y.$$

In other words,

$$[x]_{\mathcal{B}_T} = T^{-1}(\{Tx\}) \quad \text{for } x \in X.$$

We thus have by the above proposition that the conditional measure $\mu_x^{\mathcal{B}_T}$ is supported on the fiber $T^{-1}(\{Tx\})$ and only depends on the fiber. More precisely, we have a family of measures μ_y on X, $y \in Y'$ for some Borel $Y' \subset Y$ such that μ_y is supported on $T^{-1}(\{y\})$ and such that for bounded Borel $f : X \to \mathbb{C}$ we have that

$$x \mapsto \int_X f(u)d\mu_{Tx}(u)$$

is $\mathcal{B}_T$ measurable and

$$\int_A f(x)d\mu(x) = \int_A \int f(u)d\mu_{Tx}(u)d\mu(x) \qquad \text{for } A \in \mathcal{B}_T.$$

Now be Proposition C.4 we have that there exists $g : Y' \to \mathbb{C}$ which is Borel so that

$$g(Tx) = \int_X f(u)d\mu_{Tx}(u).$$

In particular, this means that

$$y \mapsto g(y) = \int_X f(u)d\mu_y(u)$$

is Borel on Y'.

Now for Borel $B \subset Y'$ we have that $A = T^{-1}B \in \mathcal{B}_T$ and the equation above says that

$$\begin{aligned}\int_A f(x)d\mu(x) &= \int_A g(Tx)d\mu(x) \\ &= \int_B g(y)d(T^*\mu)(y) \\ &= \int_B \left(\int_X f(u)d\mu_y(u) \right) d(T^*\mu)(y)\end{aligned}$$

which completes the proof. □

It is customary to define

$$\mu(B \mid T = y) := \mu_y(B)$$

and

$$E(f|T = y) := E(f|\mathcal{B}_T)(x)$$

for $y = Tx$. The law of total probability then says

$$\mu(A \cap T^{-1}B) = \int_B \mu(A|T = y)d\mu_Y(y)$$

for any Borel $A \subset X, B \subset Y$ where $\mu_Y = T^*\mu$.

Appendix D. Conditional Independence

In this section, fix a standard Borel probability space (Ω, μ) and suppose we have maps $X : \Omega \to \mathcal{X}$, $Y : \Omega \to \mathcal{Y}$ and $T : \Omega \to \mathcal{T}$ to some standard Borel spaces. We also let $\mathcal{A}_X, \mathcal{A}_Y, \mathcal{A}_T$ be the σ-algebras induced by these masp, respectively.

Definition D.1. We say that X and Y are conditionally independent given T if for all $A \subset \mathcal{X}, B \subset \mathcal{Y}$ we have that

$$\mu(X^{-1}A \cap Y^{-1}B|T = t) = \mu(X^{-1}A|T = t)\mu(Y^{-1}B|T = t) \quad \text{for } T^*\mu\text{-almost all } t \in \mathcal{T}.$$

In particular, if T is constant then we say that X and Y are independent, i.e., X and Y are independent if and only if for all $A \subset \mathcal{X}, B \subset \mathcal{Y}$ we have

$$\mu(X^{-1}A \cap Y^{-1}B) = \mu(X^{-1}A)\mu(Y^{-1}B).$$

Example D.2. Suppose $\Omega = [0,1] \times [0,1] \cup [1,2] \times [1,2]$ is the disjoint union of two squares and μ is the uniform probability measure on Ω. Let $X, Y : \Omega \to [0,2]$ denote the projections $X(x,y) = x, Y(x,y) = y$ and let $T : \Omega \to \{0,1\}$ be the map $T(x,y) = \mathbb{1}_{[0,1]^2}((x,y))$. We will show that X, Y are conditionally independent given T but are not independent. First, note that for $U \subset \Omega$ we have that

$$\mu(U) = \frac{1}{2}\lambda_2(U)$$

where λ_d denotes the Lebesgue measure on $\mathbb{R}^d$. It now follows that

$$\mu(U|T = 0) = \lambda_2(U \cap [0,1]^2)$$

and

$$\mu(U|T = 1) = \lambda_2(U \cap [1,2]^2).$$

Now for $A, B \subset [0,2]$ we have that

$$\begin{aligned}\mu(X^{-1}A \cap Y^{-1}B|T = 0) &= \lambda_2(A \times B \cap [0,1]^2) \\ &= \lambda_1(A \cap [0,1])\lambda_1(B \cap [0,1]) \\ &= \lambda_2(X^{-1}A \cap [0,1]^2)\lambda_2(Y^{-1}B \cap [0,1]^2) \\ &= \mu(X^{-1}A|T = 0)\mu(Y^{-1}B|T = 0)\end{aligned}$$

and a similar calculation holds for conditioning on $T = 1$, which shows conditional independence.

Independence does not since if we take $A = [0,1], B = [1,2]$ then

$$\mu(X^{-1}A \times Y^{-1}B) = \mu([0,1]^2 \cap [1,2]^2) = 0$$

but

$$\mu(X^{-1}A) = \mu([0,1]^2) = \frac{1}{2}$$

and

$$\mu(Y^{-1}B) = \mu([1,2]^2) = \frac{1}{2}.$$

Example D.3 (Independent but not conditionally)**.** Let $\Omega = [0,1]^2$ and suppose μ is the uniform measure on Ω. Let $X, Y : \Omega \to [0,1]$ be the projections. Then X and Y are independent. Let $T : \Omega \to [0,2]$ by $T(x,y) = x + y$. Then in fact X and Y are not conditionally independent given T. For example, let $A = B = [0,1/2]$. Then for $1 < t < \frac{3}{2}$ we have that $(A \times B) \cap T^{-1}(t) = \emptyset$ and so $P(A \times B | T = t) = 0$. However $P(X^{-1}A | T = t) > 0$ and $P(Y^{-1}B | T = t) > 0$.

Observe that we have a map $S : \Omega \to \mathcal{Y} \times \mathcal{T}$ given by $S(\omega) = (Y(\omega), T(\omega))$. We let

$$\mu(A | Y = y, T = t) = \mu(A | S = (y,t)).$$

Lemma D.4. We have that X and Y are conditionally independent given T if and only if for all $A \subset \mathcal{X}$ we have that

$$\mu(X^{-1}A | Y = Y\omega, T = T\omega) = \mu(X^{-1}A | T = T\omega) \quad \text{for } \mu\text{-almost all } \omega \in \Omega.$$

Proof. First assume X and Y are conditionally independent. Thus we must show that the right hand side satisfies the definition of conditional probability with respect to the random variable S. The right hand side is S measurable, as required. Thus it remains to show that

$$\int_{S^{-1}U} \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) = \mu(X^{-1}A \cap S^{-1}U)$$

for all $U \subset \mathcal{Y} \times \mathcal{T}$ and $A \subset \mathcal{X}$. By the monotone class lemma, it is enough to show it for $U = B \times C$. We have

$$\begin{aligned}
\int_{S^{-1}(B\times C)} \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) &= \int_{Y^{-1}B \cap T^{-1}C} \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_{\Omega} \mathbb{1}_{Y^{-1}B}(\omega) \mathbb{1}_{T^{-1}C}(\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_{\Omega} E(\mathbb{1}_{Y^{-1}B} \mid T = T\omega) \mathbb{1}_{T^{-1}C}(\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_{T^{-1}C} \mu(Y^{-1}B | T = T\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_{T^{-1}C} \mu(Y^{-1}B \cap X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \mu(Y^{-1}B \cap X^{-1}A \cap T^{-1}C) \\
&= \mu(X^{-1}A \cap S^{-1}(B \times C))
\end{aligned}$$

as required.

Now we prove the converse. Thus we must assume that for $A \subset \mathcal{X}$ we have

$$\mu(X^{-1}A | Y = Y\omega, T = T\omega) = \mu(X^{-1}A | T = T\omega) \quad \text{for } \mu\text{-almost all } \omega \in \Omega$$

and show that

$$\int_C \mu(X^{-1}A | T = t) \mu(Y^{-1}B | T = t) dT^*\mu(t) = \int_C \mu(X^{-1}A \cap Y^{-1}B | T = t) dT^*\mu(t)$$

for all $C \subset \mathcal{T}$. Using the law of total probability we have

$$
\begin{aligned}
\int_C \mu(X^{-1}A \cap Y^{-1}B | T = t) dT^*\mu(t) &= \mu(X^{-1}A \cap Y^{-1}B \cap T^{-1}C) \\
&= \mu(X^{-1}A \cap S^{-1}(B \times C)) \\
&= \int_{B \times C} \mu(X^{-1}A \mid Y = y, T = t) d(S^*\mu)(y,t) \\
&= \int_{Y^{-1}B \cap T^{-1}C} \mu(X^{-1}A \mid Y = Y\omega, T = T\omega) d\mu(\omega) \\
&= \int_{Y^{-1}B \cap T^{-1}C} \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_\Omega \mathbb{1}_{Y^{-1}B}(\omega) \mathbb{1}_{T^{-1}C}(\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_\Omega E(\mathbb{1}_{Y^{-1}B} | T = T\omega) \mathbb{1}_{T^{-1}C}(\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_\Omega \mu(Y^{-1}B \mid T = T\omega) \mathbb{1}_{T^{-1}C}(\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_C \mu(Y^{-1}B \mid T = t) \mu(X^{-1}A \mid T = t) dT^*\mu(t)
\end{aligned}
$$

□

Appendix E. Useful results

Lemma E.1. Suppose that ν is a σ-finite positive measure on X. Then there exists a probability measure ν' such that $\nu \ll \nu'$ and $\nu' \ll \nu$ (i.e., $\nu(A) = 0$ if and only if $\nu'(A) = 0$).

Proof. We only need to consider the case $\nu(X) = \infty$. There exists a countable partition $X_1, \ldots$ of X such that $0 < \nu(X_i) = d_i < \infty$. Now define

$$\nu'(A) = \sum_{i=1}^{\infty} 2^{-i} d_i^{-1} \nu(A \cap X_i).$$

□

Lemma E.2. Let ν be a σ-finite positive measure on X and ;et $\mathcal{P}$ be a family of probability measures on X such that $P \ll \nu$ for all $P \in \mathcal{P}$. Then there exists $a_1, a_2, \ldots \in [0,1]$ with $\sum_i a_i = 1$ and $P_1, P_2, \ldots \in \mathcal{P}$ such that

$$P \ll \sum_{i=1}^{\infty} a_i P_i \quad \text{for all } P \in \mathcal{P}.$$

Proof. By the previous Lemma, we only need to focus on the case that $\nu(X) = 1$. Let

$$\mathcal{Q} = \{\sum_{i=1}^{\infty} a_i P_i \mid \sum_i a_i = 1, a_i \geq 0, P_i \in \mathcal{P}\}$$

be the set of countable convex combinations of measures in $\mathcal{P}$. Observe that $Q \ll \nu$ for all $Q \in \mathcal{Q}$. For $Q \in \mathcal{Q}$ let $f_Q = dQ/d\nu$ be the density of Q and let

$$S_Q = \{x \in X \mid f_Q(x) > 0\}.$$

We remark that f_Q and S_Q are almost surely well defined. Now let $c = \sup_{Q\in\mathcal{Q}} \nu(S_Q) \leq 1$. Let $Q_1, Q_2, \ldots \in \mathcal{Q}$ be such that $\nu(S_Q) \to c$. Now there exists Q_0 such that

$$S_{Q_0} = \cup_{i=1}^{\infty} S_{Q_i}$$

and thus $\nu(S_{Q_0}) = c$. We claim that $P \ll Q_0$ for all $P \in \mathcal{P}$, thus completing the proof. Suppose it were not the case. Then there exists $A \subset X$ such that $P(A) > 0$ but $Q_0(A) = 0$. Now let

$$Q' = \frac{1}{2}\left(Q_0 + P\right).$$

We have that $Q_0 \ll Q'$ and so $S_{Q_0} \subset S_{Q'}$. Since $Q_0(A) = 0$ we have that $\nu(A \cap S_{Q_0}) = 0$. On the other hand

$$0 < P(A) = \int_A f_P d\nu \leq 2 \int_A f_{Q'} d\nu$$

and so this means that there is $A' \subset A$ with $\nu(A') > 0$ and $f_{Q'}(x) > 0$ for $x \in A'$. Thus

$$S_{Q_0} \cup A' \subset S_{Q'}$$

and so

$$\nu(S_{Q'}) \geq \nu(S_{Q_0}) + \nu(A') > c,$$

which is a contradiction. $\square$